The University of Melbourne
Semester Two 2022

MULT20015

Elements of Quantum Computing
Final Exam

3pm, November 7, 2022.

Reading Time:	**15 minutes**
Exam Duration:	**2 hours**
This paper has:	**5 pages (including this front page)**

Instructions to Students:

Attempt all 7 questions.
Total marks: 100

Please start each new question on a new page in your script book. Write the question number on each page.

Allowed calculator model: Casio FX82 (any suffix)

This exam paper is to remain in the examination room.

Instructions to invigilators: this exam paper is to remain in the examination room.

Question 1 [3 + 4 + 5 = 12 marks]

Consider a qubit state:

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle$$

with amplitudes $a_0 = 0.543 - i\,0.747$ and $a_1 = i\,0.383$.

(a) Plot the amplitudes a_0 and a_1 in the complex plane.

(b) Convert the state to polar notation form:

$$|\psi\rangle = |a_0|e^{i\theta_0}|0\rangle + |a_1|e^{i\theta_1}|1\rangle,$$

with the basis state phase angles θ_0 and θ_1 in radians (expressed as multiples of π).

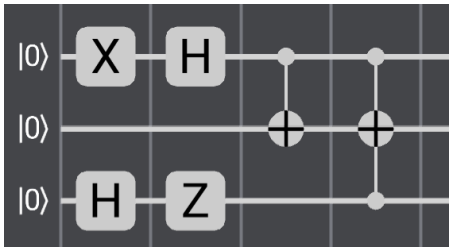
(c) Now convert to "Bloch sphere" form:

$$|\psi\rangle = e^{i\theta_{\text{global}}}\left(\cos\frac{\theta_{\text{B}}}{2}|0\rangle + \sin\frac{\theta_{\text{B}}}{2}e^{i\phi_B}|1\rangle\right),$$

specifying the global phase θ_{global} and Bloch angles θ_{B} and ϕ_B in radians (multiples of π). Hence, plot on the Bloch sphere.

Question 2 [5 + 3 + 3 = 11 marks]

Consider the following circuit comprising three qubits and four time steps:



(a) Write down the state of the three-qubit system at each of the different time-points. Show your working.

(b) If after this circuit a measurement is conducted on the top qubit and produces the result "0", what is the resulting state of the system? Is the resulting state entangled? Explain.

(c) If instead a measurement is conducted on the bottom qubit and produces the result "0", what is the resulting state of the system? Is the resulting state entangled? Explain.

Question 3 [3 + 2 + 3 + 3 = 11 marks]

Alice and Bob would like to establish a secret key quantum key distribution (QKD) using qubit states encoded in the y and z bases (i.e. different to the usual basis choice described in lectures).

Additional information:

Z basis states: $|0\rangle$, $|1\rangle$

Y basis states: $|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$.

The operator to transform from the Z basis to the Y basis has a matrix representation of $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$.

The operator to transform from the Y basis to the Z basis has a matrix representation of $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix}$.

Assume that all qubits are initially prepared in a $|0\rangle$ state.

(a) If Alice sends a qubit in the y-basis, and would like to send the bit 1, which quantum state could she send?

(b) If Eve, the eavesdropper chooses to measure this qubit in the z-basis, what are her possible measurement probabilities and results?

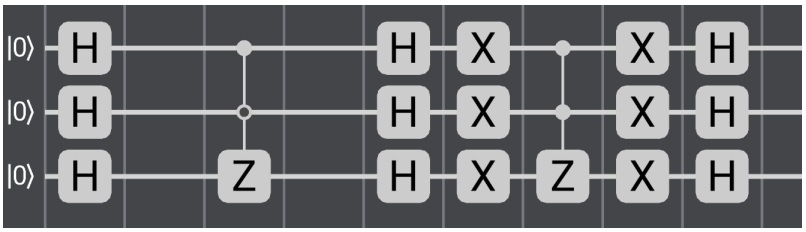
(c) Eve sends on the qubit in the z-basis. Bob chooses to measure in the y-basis. What are his possible measurement results, and probabilities?

(d) Bob happens to measure "0". How might this let Alice and Bob know that there is an eavesdropper on the line?

Question 4 [5 + 4 + 6 = 15 marks]

(a) Briefly describe the main parts of Grover's algorithm and how quantum search works.

(b) A circuit comprising an initial iteration of Grover's algorithm is shown below.



Reproducing the circuit in your script book, explain what each stage of the algorithm does. In this case, which state is marked?

(c) For a four qubit system, design an oracle which marks (i.e. applies a -1 phase) the numbers which are powers of 2 less than 16. You may use multiply controlled gates.

Question 5 [3 + 3 + 4 + 6 = 16 marks]

Consider functions with 3 bits as input and 1 bit as output.

(a) How many unique functions of this type exist?

Consider the following collection of functions:

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Minimum:   F_MIN {000,001,010,011,100,101,110,111} -> {0,0,0,0,0,0,0,1}
Maximum:   F_MAX {000,001,010,011,100,101,110,111} -> {0,1,1,1,1,1,1,1}
Majority:  F_MAJ {000,001,010,011,100,101,110,111} -> {0,0,0,1,0,1,1,1}
Parity:    F_PAR {000,001,010,011,100,101,110,111} -> {0,1,1,0,1,0,0,1}
Single:    F_SIN {000,001,010,011,100,101,110,111} -> {0,1,1,0,1,0,0,0}
Double:    F_DOU {000,001,010,011,100,101,110,111} -> {0,0,0,1,0,1,1,0}
Symmetric: F_SYM {000,001,010,011,100,101,110,111} -> {1,0,1,0,0,1,0,1}
Even:      F_EVE {000,001,010,011,100,101,110,111} -> {1,0,1,0,1,0,1,0}
Odd:       F_ODD {000,001,010,011,100,101,110,111} -> {0,1,0,1,0,1,0,1}
Zero:      F_ZER {000,001,010,011,100,101,110,111} -> {0,0,0,0,0,0,0,0}
One:       F_ONE {000,001,010,011,100,101,110,111} -> {1,1,1,1,1,1,1,1}
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(b) Within the collection written above, there exists functions that are constant, balanced, or neither. Describe what it means for a function to be constant or balanced. For each function of the collection, write down if it is constant, balanced or neither.

(c) Suppose a random function is chosen from the collection which is constant or balanced and given to you as an oracle. Describe a classical algorithm to determine whether the selected function was constant or balanced. What is the cost of your classical algorithm for n = 3 inputs.

(d) Describe a quantum algorithm to determine whether the selected function was constant or balanced with just one query. Give the quantum circuit and demonstrate its working by evaluating the circuit at each time step for the case where the oracle is F_{EVE}.

Question 6 [4 + 4 + 5 + 4 = 17 marks]

(a) Given an RSA public address (n, e), determine how many unique and valid "e" can be defined. You should omit the trivial public keys (n, 1) and (n , n-1).

(b) While using the RSA modulus of (n=221, e=5), an adversary discovers that 118^2 is congruent to 1 mod n. Can you find the factors of n using this fact? If yes, show the factors with your workings.

(c) Sketch the steps involved in determining the period using Shor's algorithm for factoring an RSA modulus n=187 with a base a =2. Estimate the sizes of both upper and lower registers.

(d) Explain in one paragraph how Shor's quantum factoring algorithm poses a threat to conventional RSA cryptography.

Question 7 [4 + 8 + 6 = 18 marks]

Consider a two asset portfolio optimization problem where the expected returns of assets A and B are 2% and 6%, respectively, and their covariance matrix (expressed as % squared) is as shown below. The risk preference, budget and budget penalty are 10, \$4, and 10, respectively. The translation of this problem to QAOA is via introducing binary variables to denote a dollar invested in an asset.

	Asset A	Asset B
Asset A	0.8	-0.1
Asset B	-0.1	0.6

The expression to optimise in the portfolio optimisation problem is:

$$\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$$

subject to $\sum_{i=1}^{n} w_i = 1$.

(a) In your script book, complete the table below by calculating the return, risk and the value of the portfolio for the given weights in the first column (denoted 'Weights').

Weights	Portfolio return	Portfolio risk	Portfolio value
$w_A = 0.3$, $w_B = 0.7$			
$w_A = 0.25$, $w_B = 0.75$			

(b) The full Hamiltonian for this problem is given by $H = H_1 + H_2$. For the purposes of this part of the question, we will consider only H_1, given by the expression below:

$$H_1 = 9\, Z_1 Z_4 + 4.5\, Z_1 Z_6 + 8\, Z_5 Z_6 + 13 Z_1,$$

where $Z_i \in \{-1, +1\}$. Let the value of parameters used in QAOA be $\alpha = -0.1, \beta = 0.2$. Draw the QAOA circuit to prepare the trial state corresponding to H_1 only. Clearly indicate the involved qubits and the angles of rotation in the various gates.

(c) Imagine that you have carried out an iteration of the QAOA algorithm, using the full Hamiltonian, H, of the problem which provided $|01111000\rangle$ as the state with the highest probability. Here the first 4 qubits correspond to Asset A, and the second 4 qubits correspond to Asset B. Compute the weights of Assets A and B implied by this state. Does the quantum state imply an optimal solution to the given portfolio optimisation problem? If the answer is yes, discuss if the quantum state is unique. If the answer is no, discuss which parameters you have to change to get an optimal state.

Additional information:

Expected portfolio return: $\sum_{i=1}^{n} w_i r_i$

Expected portfolio variance: $\sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$ where $\sigma_{ii} = \sigma_i^2$

End of Examination